Orbit-Evidence

Relational Validity Checks for Learning-Assisted Satellite Communication Experiments

A chronological split constrains order inside one dataset.

Some deployment-validity conditions are not in that dataset.

Anonymous Submission

Ordering is not deployment validity

Learning components now sit inside satellite communication software: correcting a propagated orbit, adapting a link parameter, gating a learned branch.

Because these systems are temporal, we evaluate with a chronological split. That is necessary, and the conditions under which it is valid are well established.

The claim of this paper

It is not sufficient — and the gap is one of *kind*, not degree.

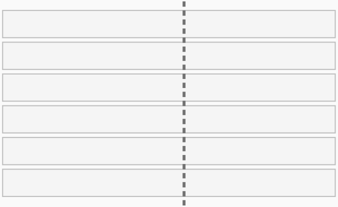
A passing split can coexist with a deployment claim that was never checkable from the data in hand.

Train on the past, test on the future — and still be unable to say the experiment was valid.

Four obligations a split cannot reach

one realised dataset

train test



a chronological split is a predicate over these rows

outside →

Availability
past-dated but unpublished

Hidden state
what is not a column

Row membership
which rows exist at all

Statistical unit
is the unit exchangeable

Each of the four has a counterexample that is not in the table the split examines.

The pivot: some validity is relational

across executions

run A ⟷ run B

same manifest, different output
⇒ the manifest is incomplete

across source states

source at t_d ⟷ source later

a feature past-dated here was
not yet retrievable there

across aggregation levels

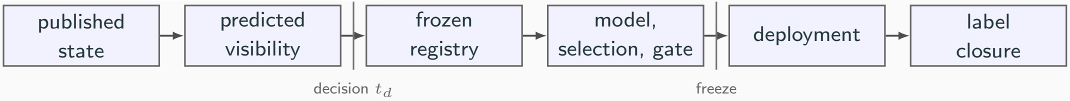
declared unit ⟷ coarser level

dependence at the coarser level
⇒ the unit is too fine

Row-local *validity is decidable from one realised run.* **Relational** *validity needs a second execution, a second source state, or a second aggregation level.*

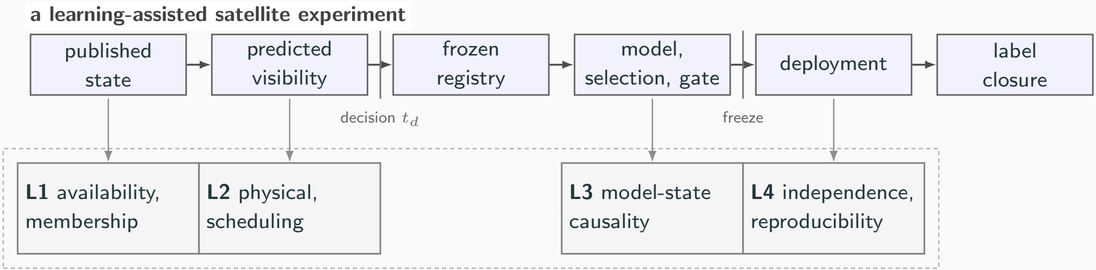
Orbit-Evidence: validity as an executable contract

a learning-assisted satellite experiment



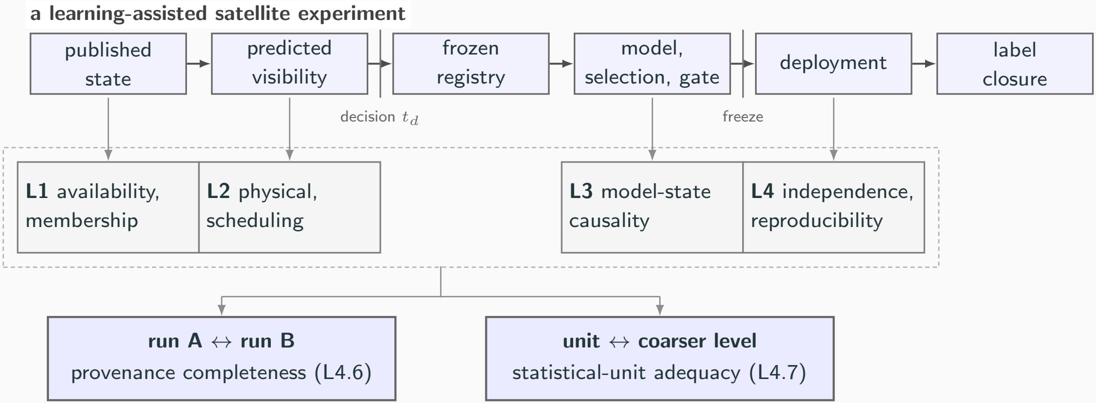
The experiment acts at t_d from state it already holds.

Orbit-Evidence: validity as an executable contract



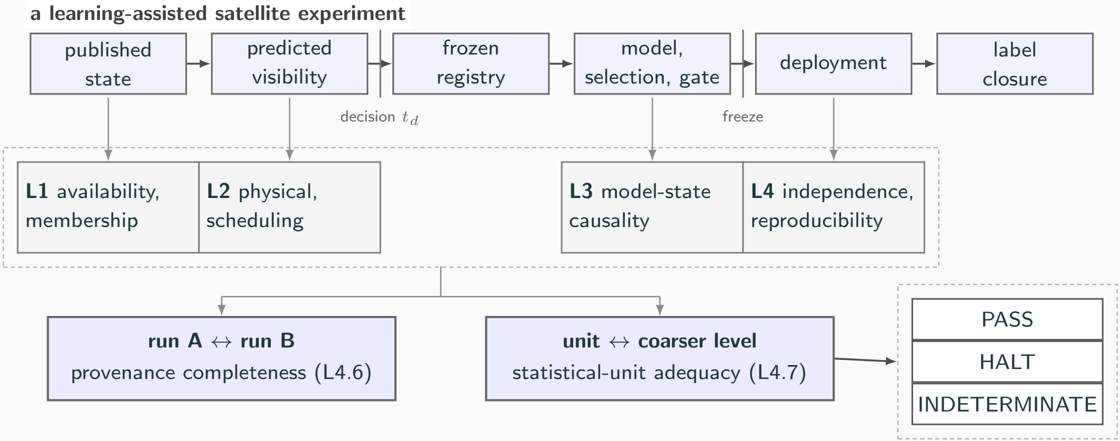
19 executable rules in four layers observe it — most inspect a single execution.

Orbit-Evidence: validity as an executable contract



Two cannot: the relational checks compare executions, or aggregation levels.

Orbit-Evidence: validity as an executable contract



A design that cannot support a decision returns INDETERMINATE — *never* PASS.

The two relational checks

L4.6 — differential provenance

Two runs, identical manifest hash, different output $\Rightarrow$ the manifest is incomplete.

Falsifies incompleteness *without enumerating* the dependencies a manifest might omit. One-sided by construction: failure is a counterexample, success is not a certificate.

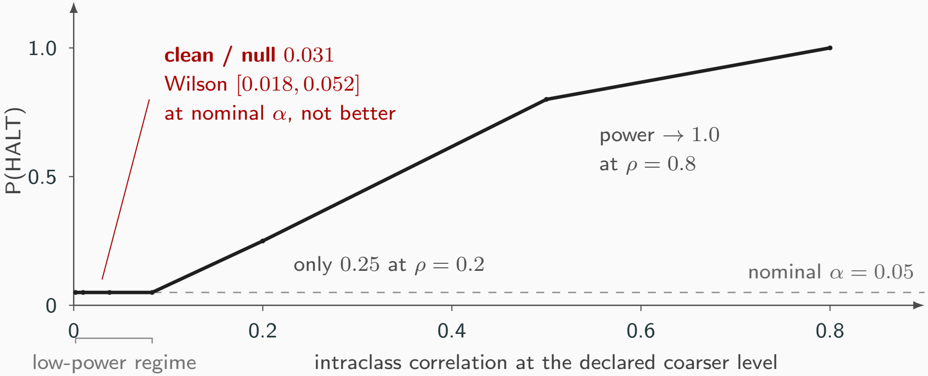
L4.7 — statistical-unit adequacy

Does dependence survive one declared level coarser? Referenced to a permutation null.

And it *abstains*: below four coarser groups the smallest reachable p is $1/15 = 0.067 > \alpha$, so no such design could reject at all. That is an *attainability* floor, not a power floor.

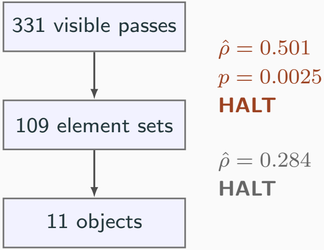
One judgement generates both: neither obligation can be certified, so we accept checks that can only refuse.

Calibrating the gate before trusting it



14 halts in 450 clean evaluations. The interval, not the point, is the claim — and it contains α.

Does the unit problem exist in real catalogue data?



Observable: the *in-track update increment* between consecutive fits.

What this means

Exchangeability of samples sharing a generating element set is rejected — one-sided lower bound 0.376. The element set is not exchangeable either: *no level tested here is*.

What it does not mean

- Not truth error — an *update increment* between fits sharing an observation arc, median $0.226\,$km.
- Not a universal unit: the rule adjudicates the level it is *given*.
- Not every observable: on elevation the same rule returned $\hat{\rho} = 0.000$.

Does the frozen contract work outside this project?

A public spacecraft-telemetry LSTM artifact, frozen at a commit chosen *before* inspection. Detector `sha256` unchanged throughout.

5 PASS	3 HALT	1 INDET	5 N/A	5 N/OBS
rule verdicts			applicability dispositions	

5 of 19 recorded *not observable* rather than passed — never scored as compliance.

Pre-registered intervention on the L4.1 halt

Correcting *only* the overlap: **HALT** $\rightarrow$ **PASS**, and the selected checkpoint changes in 5 of 5 paired seeds — reproducible bit-for-bit.

The downstream metric moved on three seeds, *in both directions*. Not a null: the smallest attainable two-sided p is $0.0625 > \alpha$.

$\Rightarrow$ **not estimable** at the available paired-run resolution — a disposition of *this study's* evidence, not a rule verdict.

No claim about the upstream paper's validity, nor about improving anomaly detection.

What is actually new

Not the primitives.

ICC(1), permutation inference, metamorphic relations, hermetic build closure, mutation testing — all established, none proposed here.

Not a checklist.

The 19 rules are the satellite *instantiation*, not the contribution.

The method is the conversion

1. Identify which validity obligations are *relational*.
2. Choose the counterfactual execution or coarser level that makes each *falsifiable*.
3. Encode the invariant as a contract — and let an undecidable case return INDETERMINATE rather than pass.

A permutation reference is ordinarily an analysis step. Here it calibrates a *gate* — with an operating characteristic, and an abstention state.

What this does not establish

- **No completeness claim.** L4.6 is bounded by the caller's variation set; the input nobody thought of is absent by construction.
- **No RF or packet-level result.** The object of study is the experiment, not the link.
- **No improvement in learned accuracy.** The contract refuses claims; it does not make models better.
- **Curated faults are regression evidence.** Faults and rules largely share an origin, so 17/17 measures *reachability*; 16 of 19 rules have a demonstrated red fixture.
- **One third-party artifact, one channel.** It partially breaks the loop. It does not close it.
- **PASS means not rejected**, never verified.

Stated because a validity paper that overstates its own evidence has refuted itself.

Chronological separation remains necessary.
It is not sufficient.

Some assumptions must become **executable**, **falsifiable**,

and allowed to **refuse** an unsupported claim.

Backup — why the abstention floor is not a power floor

Three coarser groups of two units admit only $6!/(2!^3 3!) = 15$ distinct assignments. The smallest reachable p is $1/15 = 0.067 > \alpha$: *no* design that small can reject at the nominal level, at any effect size.

Four groups of two admit 105, whose exhaustive reference reaches 0.0095.

So INDETERMINATE is a statement about the *design's resolution*, not about the strength of the effect. Collapsing it into PASS is how a design that could not have rejected comes to look like one that did not.

Backup — reproduction and provenance

- `make gate` regenerates the fault matrix and L4.7's operating characteristic, and *refuses to build* if a headline number disagrees with the artifact at the sentence claiming it. 64 numbers are bound this way — including every number on these slides.
- The two external studies are hash-verified rather than regenerated: their inputs do not ship.
- `make external-consequence-verify` checks the committed third-party results against the frozen commit, the frozen detector hash and the recorded data hashes — 16 checks, no training.
- The telemetry arrays came from a *checksum-verified mirror* after the upstream endpoint became unavailable; per-file hashes matched two independently published checksum sources. That is mirror concordance, *not* verification by the original publisher.